

Impact of Uncertainties in Power Demand Estimation on the Optimal Design of Renewable Energy Sources and Storage Systems

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Abstract—Load forecasting models are widely used to examine future electricity system. Typically, to estimate the actual and future electricity demand, these models constitute a database from different sources. Since these data are heterogeneous, the electricity demand pattern will be subjected to high uncertainties. Therefore, the effective design of renewable energy systems will be exposed to these uncertainties. For this reason, it is important to understand how the changes in the electricity demand profiles affect the modeling results in terms of system cost and electricity production mix. This paper presents a demonstration of the impact of the uncertainties on the methodology of distributed renewable energy sources and battery storage systems selection and design. This study starts with an uncertainty characterization to describe the input patterns of the energy planning model. Then, an uncertainty analysis is made to examine the model's output variation given the uncertain patterns. A detailed case study is conducted for a distribution network. The uncertainties in the demand patterns has shown a stronger impact on the electricity supply mix than on the installed storage capacities and system costs that have shown a consistent behavior.

Keywords—electricity demand pattern, uncertainties, renewable energy systems, battery storage systems, distribution network

I. INTRODUCTION

Long-term planning of energy supply-demand is necessary to meet sustainable development requirements of a country. Accurate estimations can help decision makers know the size and trend of future energy consumption to better plan the future design and operation of the production and distribution system [1]. The electricity sector shows a considerable increase in demand [2], this can be seen as an opportunity for Renewable Energy Sources (RES) integration to prevent high CO₂ emissions from fossil fuels. Modeling of electricity consumption is of capital importance to electricity network planning and operation. Different models were used in the literature for load forecasting [3], [4]. They fall into two broad groups : bottom-up [5] and top-down models [6]. Since these models are generally designed to estimate future energy demand, they rely on long-term expectations for important parameters. Therefore, they are likely to give low precision and accuracy of values, often underestimating or overestimating [7].

Uncertainty in demand side estimation introduces additional complexity alongside network design decision including RES sizing and placement, Battery Energy Storage (BES) dimensions, etc. This can influence the design process that combines technical, economic and policy aspects. Under such conditions, it is crucial to understand how uncertainty influences the optimal design and to identify the most influential uncertain parameters [8]. To address these concerns, Uncertainty Analysis (UA) must be applied. UA studies the variability of a model's output given uncertain input parameters. Monte Carlos (MC) simulations are performed in [9] for the optimal dispatch of an energy hub system under uncertain prices. Moreover, UA of energy demand has been investigated in different works, Li et al. [10] used MC simulations to study the effect of uncertainties in energy demand in the design of trigeneration system. The same method is used in [11] to study uncertainties in energy demand and solar radiation for the design of an off-grid energy system. A four-step framework for the investigation of uncertainties in optimal design of distributed energy system was developed in [8]. In the previous works [8], [10], [11], the UA is studied by varying the input parameters that characterize the demand curves (surfaces, temperatures, etc.) and they have a special focus on buildings. Overall, the number of studies performing the impacts of demand pattern on systems that include electricity distribution or transmission remains limited. Among them, a study of the effect of electricity demand on country scale electricity system was performed in [12]. In this study, different seasonal typical demand profiles were tested and their effect on system cost and electricity supply mix and storage capacities is proved.

Characterizing the electricity demand patterns extracted from load forecasting models in a blind way for the distribution network was not discussed in the literature. That means characterizing the demand patterns without having the possibility to manipulate and variate the input data of the load forecasting models. Motivated by this observation, this paper intends to develop a UA model to address the impact of profiles uncertainties extracted from a load forecasting model on total electricity cost and electricity generation mix in a distribution network.

II. METHODOLOGY AND APPLICATION

A. Electric Demand Curve Modeling

This work is intended to the electrical distribution system, a load forecasting model that use the same approach developed by Barbier et al. [13] was adopted to simulate electric load curves applied to the electrical distribution network. These curves are generated using a bottom-up approach as detailed in [14]. In such approaches, the input parameters are macroscopic data coming from large databases (e.g. national databases, Distribution System Operators (DSO)'s client information, temperature and irradiation data...). These databases are firstly matched. Then, a simulation of the operation and the consumption behavior is performed as detailed in [13].

The input parameters of the database are heterogeneous since they are extracted from different sources. This results in the existence of missing values in the matched database. In this case, missing data is replaced by values that follow the probabilistic distribution of the input data. Therefore, repeating the same calculation several times without fixing this randomness will result in different curves.

B. Uncertainty Characterization of Energy Demand Patterns

The purpose from the Uncertainty Characterization (UC) is to identify the model's uncertain parameters and to assign of a mathematical description to their uncertainty. The most common methods used to describe parameters uncertainties are the probabilistic methods that treats parameters as random variables following probability density functions [15].

In this work, a theoretical case study is built for the article, the chosen topology is a Medium-voltage rural distribution benchmark network showed in Figure 3 [16]. The input data for the electricity demand model are randomly selected from the database. Thus, 25 simulations covering one year are generated for each demand bus of Figure 3. The samples of the simulation results for a substation are depicted in Figure 1.

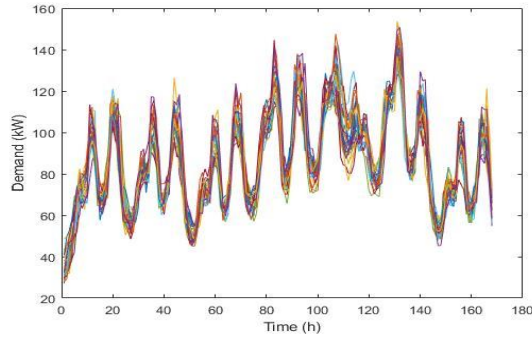


Fig. 1. Simulation results example for one substation.

Different tests were performed to select the most appropriate probability law that describes the patterns uncertainties. Firstly, a sample of points was tested with 15 tests. Secondly, 4 tests were selected according to their higher P values compared to others. The selected tests were applied to the 25 points at each time t using RStudio [17] and the results are summarized in the following Table I.

TABLE I. RESULTS OF PROBABILITY LAW TESTS ON DEMAND CURVES

Probability law	Percentage of values with ($P < 0.05$)
Normal	8.4 %
Weibull	41 %
Log normal	5.35 %
Gamma	1.38 %

Based on the results, the Gamma distribution is identified as the best uncertainty representative for the simulated demand curves. Based on these results, the alpha α and beta β characterization of Gamma law are identified for each time step of the demand time series. Then $N=100$ curves are generated using these identified parameters (α and β) at each time as shown in Figure 2.

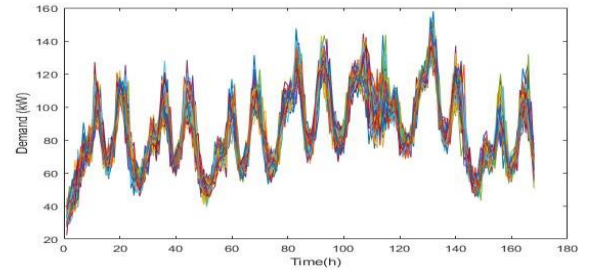


Fig. 2. Generated load patterns for a defined substation.

C. Energy Optimization Model

1) DLOPF model for RES and BES dispatch

For the task of optimal Renewable Energy Sources (RES) and Battery Energy Storage (BES) design, an Optimal Power Flow (OPF) model is first linearized as explained in [18]. Then adapted dynamically to cover a specified time period which results in a Dynamic Linear OPF (DLOPF) model. The candidate technologies are Wind Turbines (WT), photovoltaic (PV) panels and the existing Classic Generators (CG). The generated load patterns are used as input data for the simulation and the networks constraints are taken into consideration. The generated electricity is delivered to the consumers via the distribution grid. The model identifies simultaneously the optimal RES and BES design (technology sizing and placement) in addition to its optimal operating strategies over the simulated period.

The minimization of the total system cost is the main objective considered in this DLOPF model. The total system cost is expressed as the equivalent weekly cost and is projected for one year. The global objective function F_g is expressed by (1). In this equation, the total of investment and operational cost are minimized.

$$\min F_g = C_{CG} + C_{VG} + I_{WT} + I_{PV} + I_{ST} \quad (1)$$

The operational cost is the classic generators operational costs (C_{CG}) and the virtual generators operational costs (C_{VG}). Thus, the classic generator cost is calculated using equation (2) as the total produced active $P_{a,t}^{CG}$ and reactive power

$Q_{d,t}^{CG}$ multiplied by their corresponding costs, respectively $C_{CG,P}^{CG}$ and $C_{CG,Q}^{CG}$.

$$C_{CG} = \sum_{t=1}^{N_t} \sum_{d=1}^{N_G} (C_{CG,P}^{CG} \cdot P_{d,t}^{CG} + C_{CG,Q}^{CG} \cdot Q_{d,t}^{CG}) \quad (2)$$

The same concept is used for the virtual generators as expressed by equation (3). Virtual generators are supposed to be installed in all the buses to guarantee a feasible solution and to identify the power deficit in case of power need in some buses. The active and reactive virtual power are respectively $P_{i,t}^{VG}$ and $Q_{i,t}^{VG}$. Their corresponding costs are $C_{VG,P}$ and $C_{VG,Q}$.

$$C_{VG} = \sum_{t=1}^{N_t} \sum_{i=1}^{N_p} (C_{VG,P}^{VG} \cdot P_{vg}(i, t) + C_{VG,Q}^{VG} \cdot Q_{vg}(i, t)) \quad (3)$$

Equation (4) expresses the PVs total investment cost I_{PV} in terms of occupied surface S_p and the equivalent investment cost for the corresponding period by dividing the Investment for one meter square of PVs (IC^{PV}) by the PV lifetime D^{PV} .

$$I_{PV} = \sum_{p=1}^{N_{PV}} (IC^{PV} / D^{PV}) \cdot S_p \quad (4)$$

The same method is applied to calculate the investment costs for both WTs (I_{WT}) and BES (I_{ST}) in equations (5) and (6). For WT, W_w is the number of installed WTs, IC^{WT} is the investment WT cost and D^{WT} is the WT's lifetime. For BES, E_b^{bmax} is the maximum stored energy in a battery, IC^{ST} is the battery cost and D^{ST} his lifetime.

$$I_{WT} = \sum_{w=1}^{N_{WT}} (IC^{WT} / D^{WT}) \cdot W_w \quad (5)$$

$$I_{ST} = \sum_{b=1}^{N_{ST}} (IC^{ST} / D^{ST}) \cdot E_b^{bmax} \quad (6)$$

Equation (7) reflects the boundaries conditions for all the above mentioned optimization variables.

$$S.t \begin{cases} 0 \leq P_{d,t}^{CG} \leq P_{d,t}^{CGmax} & \forall g \in [0, N_G] \\ 0 \leq P_{i,t}^{VG} \leq P_{i,t}^{VGmax} & \forall i \in [0, I] \\ 0 \leq Q_{d,t}^{CG} \leq Q_{d,t}^{CGmax} & \forall g \in [0, N_G] \\ 0 \leq Q_{i,t}^{VG} \leq Q_{i,t}^{VGmax} & \forall i \in [0, I] \\ 0 \leq S_p \leq S_p^{max} & \forall p \in [0, N_{PV}] \\ 0 \leq W_w \leq W_w^{max} & \forall w \in [0, N_{WT}] \end{cases} \quad (7)$$

The active power balance equation is expressed in (8). The inlet power in a bus is formed by CG active power P_i^{CG} , virtual active power P_i^{VG} , power produced from PVs P_i^{PV} and WTs P_i^{WT} in addition to the discharged power from the batteries P_i^d . This power is either injected P_i^{inj} at bus $i/i \in I$ (where I : set of buses), it is going also to satisfy the active demand $P_{i,i}$ or charge the battery $P_{c,i}$.

$$\sum_{g \in N_G} P_i^{CG} + P_i^{PV} + P_i^{WT} + P_i^{VG} + P_i^d = P_i^l + P_i^{inj} + P_i^c \quad i \in I \quad (8)$$

The reactive power balance is the same but without RES and BES as shown in eq. (9)

$$\sum_{g \in N_G} Q_i^{CG} + Q_i^{VG} = Q_i^l + Q_i^{inj} \quad i \in I \quad (9)$$

The active energy level in the batteries at time t ($E_{i,t}^b$) is calculated using equation (10) where the energy stored in the battery is updated at each time step based on the energy level from the previous one ($E_{i,t-1}^b$), the power stored $P_{i,t}^c$ or discharged $P_{i,t}^d$ from the battery in the current time step. η_c and η_d are respectively the charging and discharging efficiencies.

$$E_{i,t}^b = E_{i,t-1}^b + \eta_c \cdot P_{i,t}^c - P_{i,t}^d / \eta_d \quad (10)$$

Equations (11) and (12) represents the classic nonlinear OPF equations for the injected active (P_i) and reactive (Q_i) power respectively. G_{ij} and B_{ij} are the real and the imaginary parts of the admittance matrix.

$$P_i = \sum_{j=1}^I (v_i v_j G_{ij} \cos \theta_{ij} + v_i v_j B_{ij} \sin \theta_{ij}) \quad (11)$$

$$Q_i = - \sum_{j=1}^I (v_i v_j B_{ij} \cos \theta_{ij} - v_i v_j G_{ij} \sin \theta_{ij}) \quad (12)$$

The apparent power is bounded by a circle represented by the equation (13) where $S_{ij,max}$ is the maximum apparent power in the branches.

$$(P_{ij})^2 + (Q_{ij})^2 \leq S_{ij,max}^2 \quad i, j \in T \quad (13)$$

The expressions for branch flows are represented in equations (14) and (15) where g and b are respectively the conductance and susceptance of branch (i, j) .

$$P_{ij} = (v_i^2 - v_i v_j \cos \theta_{ij}) g_{ij} - v_i v_j b_{ij} \sin \theta_{ij} \quad (14)$$

$$Q_{ij} = -(v_i^2 - v_i v_j \cos \theta_{ij}) b_{ij} - v_i v_j g_{ij} \sin \theta_{ij} \quad (15)$$

Equations (11) - (15) are linearized as detailed in [18]. The application of this model results in a complete design and dispatch for RES (PV and WT) and BES.

III. CASE STUDY

In this theoretical case study, the chosen topology is a Medium-voltage rural distribution benchmark network showed in Figure 3. The generated load curves are used for the demand buses. The transformers at buses 1 and 12 are defined as CGs. The potential of integrating RES and BES (available areas, etc.) are shown in Table II and their corresponding costs are resumed in Table III.

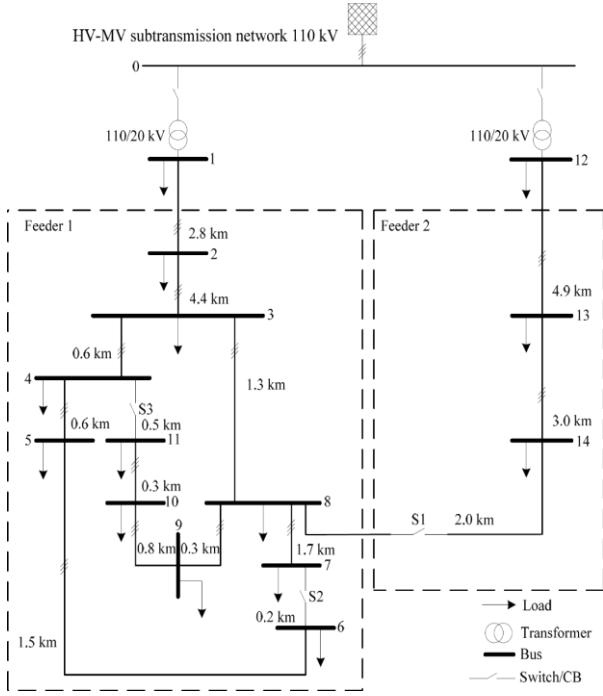


Fig. 3. Medium-voltage rural distribution benchmark network [16].

TABLE II. POSSIBLE PRODUCERS PLACEMENT AND THEIR MAXIMUM PRODUCTION CAPACITIES

Production type	Buses	Maximum capacity
PV	3,4,7,11,14	$S_{max} = 200 \text{ m}^2$ $PV_t (\text{peak load}) = 145 \text{ W/m}^2$
WT	2,3,4,5,10,13	$W_{max} = 2$ $WT_t (\text{peak load}) = 142 \text{ Kw}$
CG	1,12	2 MW
VG	all buses	10000 KW
SB	2,3,4,5,7,9,10,11,13,14	100 kWh

TABLE III. OPERATIONAL AND INVESTMENT COSTS

Production type	Operational cost (€/MWh fuel)	Investment cost
PV	0	$I_{PV} = 200 \text{ (€/m}^2)$
WT	0	$I_{WT} = 1 \text{ 300 €/KW [19]}$
CG	$C_{CG}, C_{CGQ} = 36 \text{ [20]}$	0 (already installed)
VG	$C_{VG}, C_{VGQ} = 10^{9*}$	0
SB	0	$I_{ST} = 10^3 \text{ €/KWh [21]}$

* This cost is intended to ensure that the use of virtual generators is avoided

IV. PLAN OF EXPERIENCE FOR UNCERTAINTY ANALYSIS

The examination of model output variation given uncertain input parameters is done in this work by applying the following steps:

Step 1. Definition of the load curves indicators.

The network is constituted of 14 substations. 100 curves are generated from part II.B for each substation. The following indicators are calculated for each curve:

- The total consumed energy (E_{tot}) covering the defined period (in this case one week = 168 hours).
- The peak demand in the curve (Max_E).
- The variance of the demand of the defined period (Var_E).

Step 2. For each substation, a k-means clustering is calculated based on the associated 100 curves. This clustering is based on the elements calculated in Step 1 (E_{tot} , Max_E and Var_E). 3 different clusters are resulted from the clustering in each substation.

Step 3. The grid is divided into two subgroups as shown in Figure 3. The first subgroup A covers the buses 1,2,3,4,5,6,7,8,9,10 and 11. The second subgroup B contains the buses 12,13 and 14. For each case, the demand curves of all buses in the same subgroup are extracted from the same cluster index. A curve from the selected cluster will be selected randomly.

Step 4. 9 cases are generated and summarized in Table IV. In this table, the cluster index chosen for each substation is indicated which makes a combination of demand curves for each case. DLOPF is then applied to each of them.

Step 5. For each case, the 3 output indicators are compared:

- Installed electricity supply mix.
- Installed storage capacities.
- The system cost (investment + operation).

TABLE IV. CLUSTER NUMBER ALLOCATION

	Subgroup A	Subgroup B
Case 1	1	1
Case 2	1	2
Case 3	1	3
Case 4	2	1
Case 5	2	2
Case 6	2	3
Case 7	3	1
Case 8	3	2
Case 9	3	3

V. RESULTS AND DISCUSSIONS

To evaluate the uncertainties, the obtained results from the defined cases are compared to a mean case. This latter is formed by the average demand profile from the 100*14 curves.

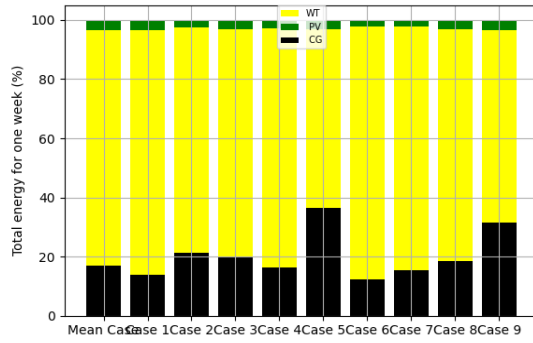


Fig. 4. Installed electricity supply mix for each case.

The optimal generation mixes are shown in Figure 4. Compared to the mean case, the CG production can vary by a maximum variation of 21% in some cases. 13% for the PV production and 4% for the WT production. The cases 1, 4 and 7 where the cluster index 1 is used for the second subgroup are showing a CG production less than the mean case where the cases in which the category 2 is chosen for the second subgroup (2, 5 and 8) are resulting in a CG production less than the mean case. This is due to the higher load in category 2 in the second subgroup where an existing CG is producing power to the 2 existing loads (13 and 14). From the Figure 5, the installed storage capacity is varying by -3.6% and $+9.5\%$ around the mean case.

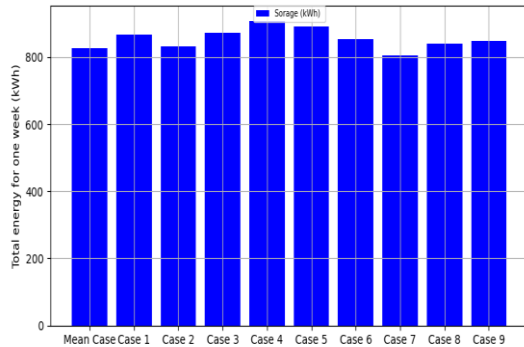


Fig. 5. Installed storage capacities for each case.

These results shows that the presence of uncertainties in electrical demand patterns has an influence on the energy mix.

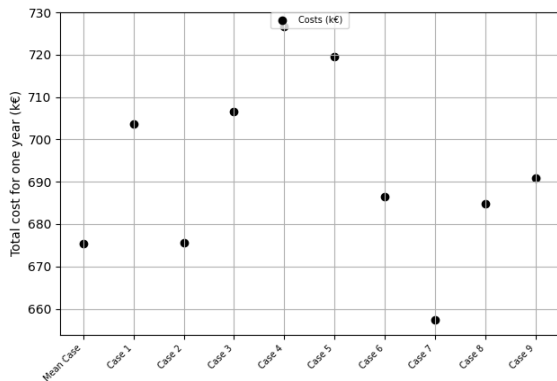


Fig. 6. System cost for each case (investment + operation).

In Figure 6, the system cost (investment + operation) of the defined cases is depicted. It is clear that, the influence of the uncertainties is projected to the system cost and is varying annually between 655 k€ and 728 k€. A difference of -2.5% and 7.85% is detected around the mean case based on the uncertainties' effect.

It can also be assumed that the variation behavior of the storage capacities is coherent with the costs. Two extreme cases are identified: the first one is case 4 which is the most expensive and the second one is case 7 (the cheapest). Between these two cases, the energy mix generated is close. The difference in costs is due to the storage capacity, which in case 4 is the largest and therefore results in a higher total cost, and in case 7 is lower, which makes the system cheaper.

Finally, it is clear that a change in demand patterns has a stronger impact on the electricity supply mix than on the system cost.

VI. CONCLUSIONS AND PERSPECTIVES

In this study, the effect of uncertainties on the electricity supply mix, storage capacity and the system cost is studied. The results show that these three indicators are influenced differently by uncertainties in demand data. The impacts of different demand patterns on installed storage capacities are in line with the system cost variation. The uncertainties in electricity demand curves have a stronger influence on the installed electricity supply mix than on system cost (21% for CGs, 13% for PVs and 4% for WTs where a variation of 10.1% is monitored for the costs). In this paper, a small number of cases is simulated based on an arbitrary division of the subgroups and an attribution of demand clusters. A detailed load pattern characterization and a plan of experience for the definition of the tested cases should be made to proceed with sensitivity analysis studies in the coming work.

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